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ORIGINAL PAPER

Bank Digitalization and Capital Reallocation

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Abstract
This study investigates the impact of bank digitalization on capital reallocation using the adoption of digital lending platforms and a difference-in-differences design. We find that digital transformation in banking facilitates the reallocation of capital from higher-risk to lower-risk sectors. This effect is particularly pronounced in regions with high levels of government influence and in city and rural commercial banks, which are more susceptible to government intervention. Our findings suggest that banks that implement digital platforms are better positioned to resist political pressures, as the data-driven, automated decision-making processes make it difficult to justify risky, politically motivated loans, while increased transparency limits local government influence over credit allocation. The effects of digitalization on capital reallocation are further amplified in banks with opaque information environments, weak internal control, and high credit risks. Additional analyses show that the adoption of digital lending platforms improves the timeliness of loan loss recognition, reduces bank risk, and enhances operating performance. This study contributes to the literature on digitalization by highlighting its role in improving capital allocation in the banking sector and offers insights into how technological advancements can mitigate the risks posed by political intervention and high-risk lending.

Keywords Bank digitalization · Digital lending platforms · Capital reallocation · Government influence · Bank risk-taking

Introduction

Unlike developed economies where equity financing plays a more central role, the financial systems in emerging markets rely heavily on debt financing, with companies primarily depending on bank loans to meet their capital needs (Allen et al., 2017). Commercial banks are essential in reallocating capital, directing funds from declining sectors to growing industries to promote innovation and economic growth. However, banks often face challenges in their capital allocation decisions, which can lead them to allocate credit based on factors other than credit risk (Chen & Lin, 2019).

Extant literature shows that political pressures often force banks to lend more to politically favored industries, even if these sectors are riskier, leading to inefficient resource allocation and increased financial instability (Agarwal et al., 2012; Huang & Thakor, 2024). This issue is particularly prevalent in emerging markets, where local governments frequently intervene in banks' credit decisions, pressuring them to lend to insolvent borrowers (Caballero et al., 2008; Cai et al., 2017; Tian et al., 2023). Banks often comply with local governments because it is easier and less costly for them to comply with political demands under traditional manual-dominated loan approval processes. While such practices aim to stabilize local economies and maintain employment, they distort capital allocation, increase exposure to high-risk sectors, and potentially destabilize the financial system. Addressing the issue of capital misallocation is, therefore, a key concern for both researchers and policymakers.

This study examines how bank digitalization influences capital reallocation. Bank digitalization refers to the integration of digital tools into financial operations to improve efficiency (Chen & Shen, 2024; Shang & Niu, 2023; Xie

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& Wang, 2023). While previous studies have used textual analysis of banks' annual reports to measure digitalization, such approaches often overlook the true digital capabilities of banks. Digitalization is a broad concept, and it can be difficult to pinpoint which specific elements of digitalization affect capital allocation. To address this, we focus on the adoption of digital lending platforms. These platforms use advanced technologies such as blockchain and machine learning to enhance credit assessments and automate approvals (Doumpos et al., 2023). Although these platforms initially target small and micro-enterprises, the technological improvements they embody have broader implications for the entire banking system, influencing how banks manage risk and allocate capital.[1] For example, big data analysis allows banks to evaluate risks more effectively across loan sizes, and blockchain technology ensures secure and transparent record-keeping for every transaction, from loan assessment to approval and post-lending monitoring. These technological improvements reduce the potential for human error and ensure more consistent management of loans of all sizes. In summary, while digital lending platforms may focus on smaller enterprises, the technological improvements they represent benefit all lending activities.

The adoption of digital lending platforms offers a unique setting for exploring the impact of digitalization on risk management and capital allocation. First, these platforms increase the costs of coordinating politically favored loans. Under traditional manual-dominated loan approval processes, it was easier and less costly for banks to comply with political demands. Digital lending platforms enhance transparency, making it costly for banks to yield to political pressures and harder to justify politically motivated lending decisions. Moreover, these platforms improve internal control, thus limiting banks from prioritizing short-term profitability at the expense of increasing non-performing loans (NPLs). By leveraging blockchain technology for transparent, traceable records and utilizing big data analytics for more accurate risk assessments, digital lending platforms reduce opportunities for opportunistic behavior and ensure sound lending practices. Consequently, banks may reduce their exposure to high-risk sectors. However, the shift to digital lending also presents significant challenges. Issues related to data security, regulatory compliance, and the need for banking professionals to acquire new skill sets have become key concerns (Chanias et al., 2019; Li et al., 2022). The autonomy provided by these platforms could be misused by technologically corrupt individuals, raising questions about the overall effectiveness of bank digitalization in improving capital allocation efficiency.

Using a sample of Chinese listed banks from 2010 to 2022 and a DID design, we find that digitalization leads to a reallocation of bank credit—from higher-risk to lower-risk sectors. Recognizing that government influence often exacerbates capital misallocation, our study explores the moderating role of government intervention. The effect of digitalization is more salient in areas with high levels of political pressures, while it is less noticeable in non-regional banks. In addition, banks may face internal incentives to lend to high-risk borrowers for short-term profit gains, particularly those with less transparency. Consistent with this, we find that greater transparency proxied by loan loss recognition timeliness and analyst coverage weakens the impact of digitalization on capital reallocation. Furthermore, the effects concentrate on banks with weaker internal control and higher credit risk, as measured by the proportion of NPLs and loan concentration. Additionally, our analyses show that digitalization enhances loan loss recognition timeliness, reduces bank risk, and improves operational performance, underscoring its broader benefits for the banking sector. These findings highlight how digitalization fosters more efficient and transparent capital allocation.

Our study adds to the growing literature on the impact of digitalization. While most existing research examines how digitalization transforms non-financial firms' business models, reduces reliance on trade credit, and enhances equity values and overall performance (Chen & Srinivasan, 2023; Correani et al., 2020), less attention has been given to its effects on commercial banks. Prior studies in this area have primarily focused on bank credit risk and performance (Chen & Shen, 2024; Cheng & Qu, 2020; Sheng, 2021; Zhao et al., 2022), often highlighting that digitalization increases overall lending. In contrast, our research explores how digitalization influences capital allocation across sectors with varying levels of NPLs, emphasizing the role of bank digitalization in withstanding political pressures and allocating capital toward low-risk sectors.

Moreover, while many studies measure digitalization by counting-related terms in annual reports, this approach may not fully capture a bank's true digital capabilities. General measures of digitalization also make it challenging to pinpoint which specific aspects influence bank decision-making. Our study addresses this gap by focusing on the adoption of digital lending platforms, which allows us to identify the causal impact of digitalization on bank capital allocation decisions.

This study also contributes to the literature on bank capital reallocation. Although banks are theoretically expected to shift capital from declining sectors to growing ones, various external and internal challenges often impede this process.

[1] We conducted a survey targeting loan officials at two city commercial banks in Chongqing and Xi'an, as well as one joint-stock bank in Beijing. The survey revealed that digital lending platforms are considered a representation of bank digitalization. The various technologies embedded in these platforms have a substantial impact on lending decisions.

Improving the efficiency of capital allocation is thus a key research question. Prior studies have shown that stricter capital requirements and institutional reforms, such as improved investor protections, facilitate the liquidation of NPLs and improve credit allocation (Keuschnigg & Kogler, 2022). Our study documents that digitalization mitigates government interference in lending decisions and improves risk management, thereby redirecting capital from higher-risk to lower-risk sectors.

Additionally, our findings offer new insights into how technology can improve fairness and efficiency in the banking sector. By showing how digitalization mitigates the negative effects of politically motivated lending and opportunism, our study provides valuable guidance for policymakers and stakeholders. Promoting the adoption of digital lending platforms can be an effective strategy to reduce the risks associated with politically motivated lending. Encouraging digitalization in banking improves the integrity of credit decisions and enhances internal control, reducing opportunities for opportunistic behavior. Stakeholders such as regulators and investors stand to benefit from improved monitoring of banks' risk exposure and operational performance, thanks to the increased transparency brought about by digital technologies.

Institutional Background and Literature Review

Institutional Background of the Chinese Banking System

China's banking system has experienced profound changes since the economic reforms of 1978, transitioning from a monolithic government-controlled structure to a more complex and diversified system. Initially, the system served primarily to fund state-owned enterprises (SOEs) and government projects, with the People's Bank of China (PBOC) acting as both a central and commercial bank (Allen et al., 2017; Zhu, 2021). Reforms to the banking industry established specific purpose banks or policy banks, such as the China Development Bank, to handle policy-driven investments, and restructured the Big Four state-owned commercial banks—Bank of China, China Construction Bank, Industrial and Commercial Bank of China, and Agricultural Bank of China—to operate with a greater focus on commercial banking activities, while still under significant government influence (He & Wei, 2023). The inclusion of joint-stock commercial banks, city commercial banks, and foreign banks has since added layers of competition and diversity, reflecting a significant shift toward a more market-oriented banking sector. Despite these reforms, the Chinese banking system continues to face challenges.

One of the key challenges is the continued influence of local governments on banks' lending decisions. Banks often align with local governments to allocate capital to high-risk sectors, driven by political pressures and the expectation of implicit guarantees. Local governments, aiming to stimulate economic growth, maintain employment, or stabilize local economies, often pressure banks to lend to politically favored sectors such as real estate, infrastructure, or other government-supported industries (Wang et al., 2019). These sectors may not always be creditworthy, but banks comply with these demands to maintain strong relationships with government officials, which can translate into regulatory advantages or future business opportunities.

Furthermore, banks may be more willing to lend to high-risk sectors because they anticipate that the government will step in with guarantees or bailouts if loans go bad. This belief lowers their perceived risk and encourages them to allocate more capital to sectors with higher NPLs. Traditional manual loan approval processes make it easier and less costly for banks to comply with political demands, as these processes often lack transparency and standardized oversight. This allows banks to meet government expectations with minimal scrutiny. However, this practice of redirecting capital from viable projects to politically favored ones often results in capital misallocation and higher NPLs, weakening the efficiency of the financial system (Allen et al., 2017; Cai et al., 2017; Song & Xiong, 2018).

State-owned banks, given their direct ties to the government, are particularly vulnerable to such political pressures. These banks often serve dual roles: operating as commercial entities while also fulfilling policy objectives. However, joint-stock commercial banks are not immune to these pressures. Although these banks are generally expected to operate under commercial principles, their managers may find it beneficial to align with local government goals to secure future benefits or avoid regulatory challenges (Allen et al., 2017; Zhang et al., 2016).[2] City and rural commercial banks, which have close ties to local governments and are crucial for regional development, are especially vulnerable to political intervention.

Bank Digital Transformation

Bank digitalization refers to the integration of digital technology into banking operations, designed to improve the

[2] A typical situation of government influence in bank lending can be seen in the handling of local government financing vehicles (LGFVs). These entities, often used by local governments to fund infrastructure projects, rely heavily on bank loans. Given the implicit government guarantee, banks may extend credit under less stringent conditions, reflecting the complex interplay between policy-driven lending priorities and the inherent risks of such practices.

efficiency of financial services (Xie & Wang, 2023). This transformation involves adopting technologies like blockchain, artificial intelligence (AI), all of which contribute to greater transparency, lower transaction costs, and enhanced risk management in banking operations (Nguyen et al., 2023; Razzaq & Yang, 2023). Blockchain technology, for instance, has revolutionized lending by enabling decentralized finance (DeFi) platform. These platforms eliminate the need for traditional intermediaries, making transactions secure and transparent (Saberi et al., 2019). Smart contracts enable automated lending and borrowing processes. AI and machine learning algorithms enhance credit scoring by evaluating borrower risk with broader data, including unconventional sources such as social media activity and mobile phone usage patterns (Nguyen et al., 2023).

An example of this transformation is China Construction Bank (CCB)'s launch of its Smart Lending Platform in 2014. This platform collects data from various external sources such as tax records, credit histories, and other financial information to automate loan approvals. By reducing unnecessary human intervention, the platform ensures faster and more accurate lending decisions. Additionally, the intelligent risk control system embedded in the platform strengthens internal control and enhances loan risk assessment and management. CCB has also incorporated blockchain technology into cross-border payments and supply chain finance, significantly improving the speed and transparency of transactions.

Hypothesis Development

Banks play a key role in capital allocation, directing funds from declining sectors to growing ones, which is vital for economic growth. However, banks often face frictions in their capital allocation decisions. These frictions, such as government intervention and the pursuit of short-term profits, can lead banks to allocate credit based on factors unrelated to credit risk (Agarwal et al., 2012; Huang & Thakor, 2024). This can result in excessive lending to high-risk sectors, increasing the likelihood of NPLs and financial instability. Bank digitalization helps address these issues by facilitating the reallocation of capital from sectors with high NPLs to those with lower NPLs.

First, digitalization increases the costs for banks to coordinate politically favored loans, thus reducing government influence on credit allocation. Local governments often pressure banks to lend to specific sectors, which can lead to risky lending practices (Agarwal et al., 2012). These pressures may come through formal or informal channels, urging banks to make loans that do not align with proper risk assessments (Becker & Ivashina, 2018). Such interventions are typically aimed at stabilizing local economies, maintaining jobs, and supporting tax revenues (Cai et al., 2017; Tian et al., 2023). For example, Huang and Thakor (2024) show that political pressures can drive banks to lend more to politically favored but high-risk industries, resulting in resource misallocation. The success of politically favored loans largely depends on banks' willingness to coordinate. Before digitalization, banks often complied with these demands, as manual loan approval processes made it easier and less costly to accommodate political demands.

Digitalization changes this by increasing the costs and risks of complying with political demands. By introducing automated systems, digital lending platforms reduce human discretion and introduce standardized processes based on data (Doumpos et al., 2023). These platforms ensure that lending decisions are more transparent, objective, and traceable, making it harder for banks to cater to political favoritism. Banks can explain to local governments that lending decisions are now based on data-driven processes and rely on objective criteria. In other words, banks that adopt digital platforms are better positioned to resist government interference. Moreover, digital technologies embedded in these platforms enable banks to recognize loan losses more promptly and improve external transparency by increasing the visibility of lending decisions (Li et al., 2024; Sun et al., 2024). This provides regulators and investors with a clearer view of the bank's financial health and risk exposure, reducing the ability of local government officials to influence credit allocation decisions without proper oversight. Local governments are less likely to pressure banks for politically motivated loans, as these actions are more likely to be detected and held accountable. While government influence may still exist, digitalization makes it costly and risky for banks to align with political interests. As a result, banks tend to lend to sectors with a lower proportion of NPLs and improve the overall quality of their loan portfolios.

Second, digitalization improves bank transparency and enhances internal control, curbing opportunistic behavior. In addition to political pressures, banks may face internal incentives to lend to high-risk borrowers for short-term profit gains. While this may boost earnings in the short run, it often leads to higher NPLs and financial instability. Banks with weaker internal control and less transparency are likelier to engage in such behavior (Jin et al., 2013; Mies, 2024). Digital lending platforms use technologies like blockchain and data analytics to increase transparency throughout the lending process—from credit evaluation to loan monitoring (Li et al., 2024). For instance, digitalization improves post-lending oversight by continuously tracking loans in high-risk sectors. This helps banks monitor loan performance more effectively and respond quickly to market changes. Banks can thus identify issues early and reduce the chances of risky lending practices. Moreover, using smart risk management systems in digital lending platforms improves internal control (Chen et al., 2014), making it difficult for banks to

prioritize short-term gains at the expense of long-term stability. Therefore, we put forward the first hypothesis:

Hypothesis 1 Bank digitalization facilitates the capital reallocation from sectors with higher proportion of NPLs to those with lower proportion of NPLs.

Bank digitalization is expected to improve the efficiency of capital allocation by promoting more objective, data-driven decision-making. However, the degree to which digitalization impacts capital reallocation likely varies across different regions and types of banks, especially those more vulnerable to government influence.

In regions with high levels of government intervention, banks have traditionally faced external pressures to allocate credit toward politically favored or high-risk sectors (Becker & Ivashina, 2018; Dinç, 2005; Huang & Thakor, 2024; Sapienza, 2004). To maintain favorable relationships with local authorities, banks often comply with these demands and misallocate capital toward higher-risk sectors, as manual-dominated loan approval processes offer low coordination costs and significant discretion. Adopting a digital lending platform enhances transparency and introduces standardized, automated processes, limiting the discretion of bank officials. This, in turn, raises the costs of yielding to political pressures, making it difficult for banks to justify lending to politically favored sectors.

City and rural commercial banks, which often have closer ties to local governments and play a key role in regional development, are particularly susceptible to political intervention (Sun et al., 2013). These banks are under more significant pressure to align their lending with local policy objectives, often leading to inefficient capital allocation. Digital lending platforms reduce the scope for human discretion, making it more difficult for these banks to favor politically connected borrowers over lower-risk ones. Therefore, bank digitalization's impact on capital reallocation is expected to be more pronounced in city and rural commercial banks as they move away from politically motivated lending toward more risk-based decision-making. Thus, we propose the following hypothesis:

Hypothesis 2 The impact of bank digitalization on capital reallocation is more pronounced in regions with high levels of government influence, and in city and rural commercial banks.

We further explore how banks with opaque information environments and weak internal control can benefit from reduced capital misallocation through digitalization.

Opaqueness provides opportunities for banks to engage in capital misallocation because it hinders the ability of regulators, investors, and auditors to monitor and assess the bank's lending practices. Digitalization increases bank transparency by making processes more visible and accountable (Li et al., 2024). Utilizing technologies like blockchain, digital systems record transactions, approvals, and decisions in a traceable way and provide secure and unchangeable records of financial transactions. This increased transparency makes it easier for auditors and regulators to monitor banking activities, ensuring the information provided is accurate and timely. By utilizing data analytics, banks can quickly identify loan losses (Sun et al., 2024), providing regulators and investors with a clearer view of the bank's financial health and risk exposure. In banks with opaque information environments, digitalization leads to greater transparency and accountability.

Additionally, digital transformation helps banks enhance internal control and improve risk management. Banks with weak internal control often face difficulties properly managing risks (Jin et al., 2013). Common issues include inadequate oversight, failure to comply with policies, and weak checks and balances (Altamuro & Beatty, 2010). These weaknesses can lead to risky behavior, with banks prioritizing short-term profits over prudent credit risk management, resulting in inefficient capital allocation.

Digitalization improves internal control by automating processes and applying tools like data analytics and machine learning. These technologies help banks improve their risk management and reduce the likelihood of human error or manipulation. Using the COSO framework, digital lending platforms improve internal control across governance, risk assessment, control activities, information sharing, and monitoring. Automated systems ensure consistent adherence to loan policies, minimizing individual discretion. Regarding risk assessment, digital lending platforms analyze multiple data sources, allowing banks to identify high-risk borrowers early and adjust capital allocation accordingly. These platforms also automate key loan processes, enhance the quality of information through real-time monitoring, and provide continuous oversight throughout the loan lifecycle (Doumpos et al., 2023). Therefore, banks with weaker internal control experience more significant improvements in capital reallocation, moving funds from higher-risk to lower-risk sectors. Thus, we propose the following hypothesis:

Hypothesis 3 The impact of bank digitalization on capital reallocation is more pronounced in banks with more opaque information environments and weaker internal control.

Research Design

Sample

Our sample includes 42 listed banks from 2010 to 2022, comprising 6 state-owned large commercial banks, 9

joint-stock commercial banks, 17 city commercial banks, and 10 rural commercial banks. The period starting in 2010 aligns with the beginning of the digital transformation of Chinese banks, which started in the mid-2010s. Our final dataset comprises 10,900 bank-year-industry observations. Banks' financial information was obtained from the Wind, CNRDS, and CSMAR databases. We collect industry-level non-performing loan data from the annual reports of the China Banking Regulatory Commission.

Key Variables

Dependent Variable

To measure the banks' capital allocation, we use the proportion of the loan a bank assigns to different sectors (*IndProp*) as a key metric. By comparing the distribution of a bank's lending across various sectors, we investigate how banks allocate funds among various industries, reflecting capital redistribution.

Independent Variables

Banks' digital transformation is a comprehensive process, and accurately measuring their digital transformation is highly challenging. Extant literature has measured corporate digitalization in three ways: indicator selection, text analysis, and questionnaire survey. However, these approaches may not fully capture a bank's actual digital capabilities. Using broad measures of digitalization also makes it challenging to determine which specific aspects influence bank capital allocation. This paper uses banks' adoption of digital lending platforms as a proxy. The digital lending platform is an advanced credit service system that utilizes technologies like big data, AI, and machine learning to digitize the entire lending process (Doumpos et al., 2023). It includes comprehensive credit evaluations, automated credit approvals, and accurate risk management, improving the efficiency and security of credit operations. The introduction of digital lending platforms marks a key stage in the digital transformation of banks, particularly in enhancing credit decision-making and risk control. $Digitalization_{i,t}$ equals one for a specific bank i in a given year t if that bank has adopted the digital lending platform by that year.

In robustness tests, we utilize the bank digital transformation index compiled by Xie and Wang (2023) from the Peking University Institute of Digital Finance (Chen & Shen, 2024; Shang & Niu, 2023). They identify the degree of bank digitalization from strategy, operation, and management perspectives. Specifically, strategic digitalization focuses on the top-level design and strategic deployment of commercial banks' digital transformation, constructed by calculating digital technology-related keywords in commercial bank annual reports; operational digitalization focuses on the integration of digital technology with the financial service provision of commercial banks, emphasizing the level of innovation in financial products, online channels, and online services; management digitalization primarily focuses on the integration of digital technology into the governance structure and organizational management of commercial banks, characterizing the level of digitalization from dimensions such as digital talents and digital cooperation.

Control Variables

Following Wang et al. (2024), we control for a series of bank financial characteristics that may affect bank efficiency, including bank size (*Size*), return on net assets (*ROA*), leverage ratio (*Lev*), loan growth rate (*Growth*), the proportion of non-performing loan (*NPL*), the proportion of loan loss provisions (*LLP*), capital adequacy ratio (*Tier1*), and loan-to-deposit ratio (*SLR*). Appendix Table 8 presents detailed variable definitions.

Model Specification

We utilize a DID design to examine the association between bank digitalization and capital reallocation.

$$\begin{aligned} IndProp_{i,j,t+1} = \alpha_0 + \alpha_1 \times Digitalization_{i,t} \times IndNPL_{j,t} \\ + \alpha_2 \times Digitalization_{i,t} \\ + \alpha_3 \times IndNPL_{j,t} \\ + \beta \times Controls_{i,t} + Bank\,FE \\ + Industry \times Year\,FE + e_{i,j,t}, \end{aligned} \tag{1}$$

where i, j, and t represent bank i, industry j, and year t, respectively. $IndProp_{i,j,t+1}$ represents the proportion of loans bank i issued to industry j in year $t+1$. $Digitalization_{i,t}$ equals one for a specific bank i in a given year t if that bank has adopted the digital lending platform by that year. We use an *ex-ante* measure of industry NPLs to divide the sample into two groups. $IndNPL_{j,t}$ is assigned a value of one if the NPL ratio for industry j exceeds the NPL ratio in 2010, and zero otherwise. The key variable of interest is the interaction term $Digitalization_{i,t} \times IndNPL_{j,t}$. A negative coefficient of $Digitalization_{i,t} \times IndNPL_{j,t}$ suggests that compared to banks without a digital lending platform, banks that have implemented a digital lending platform tend to reduce their credit allocation to industries with high NPLs. We incorporate bank fixed effects to control for the impact of time-invariant bank-level characteristics and Industry × Year Fixed Effects to isolate the supply-side effect from the demand-side influence (Hung et al., 2024; Li et al., 2023).

Table 1 Descriptive Statistics

Panel A: Non-performing loan ratio by industry

Industries	Industry-level proportion of NPLs in 2010
Accommodation and food services	3.01
Agriculture, forestry, animal husbandry, and fishery	3.13
Construction industry	0.77
Culture, sports, and entertainment industry	1.76
Education	1.64
Electricity, heat, gas, and water production and supply	1.19
Financial industry	0.3
Health and social work	1.03
Information transmission, computer services, and software industry	1.93
Leasing and business services	0.73
Manufacturing industry	1.87
Mining industry	0.25
Real estate industry	1.26
Residential services, and repair services	1.29
Scientific research and technical services	1.88
Transportation, storage, and postal services	0.97
Water conservancy, environment, and public facilities management	0.39
Wholesale and retail trade	1.56

Panel B: Summary statistics

Variables	*N*	Mean	SD	p25	Median	p75
IndProp	10,900	0.050	0.084	0	0.012	0.069
IndNPL	10,900	0.500	0.500	0	0.500	1
Digitization	10,900	0.389	0.488	0	0	1
Size	10,900	26.473	5.575	25.865	27.124	28.860
ROA	10,900	0.009	0.003	0.007	0.009	0.010
Lev	10,900	0.893	0.179	0.919	0.928	0.938
NPL	10,900	0.012	0.005	0.009	0.012	0.015
LLP	10,900	0.031	0.008	0.025	0.030	0.036
Growth	10,900	0.165	0.069	0.121	0.156	0.201
Tier1	10,900	0.106	0.024	0.096	0.106	0.119
SLR	10,900	0.714	0.146	0.636	0.713	0.787

This table presents the descriptive statistics. Panel A displays the industry-level non-performing loan ratio in 2010. Panel B provides the summary statistics for our key variables

Empirical Results

Descriptive Statistics

Panel A of Table 1 presents the average NPL ratios for 18 industries in 2010. The sectors with the highest proportions of NPLs are agriculture, accommodation and food services, information transmission, computer services and software, scientific research and technical services, and manufacturing. These sectors encompass both large and small enterprises. Panel B shows the statistics of main variables. The average of *IndProp* is 0.050. *IndNPL* varies notably with a standard deviation of 0.500, indicating diverse lending practices and risk exposure. *Digitalization* has a mean of 0.389 and a standard deviation of 0.488.

In Appendix Table 9, we provide a list of the 42 sample banks and the years they launched their digital lending platforms. The launch year is based on the first disclosure of the platform in the banks' annual reports, which are cross-checked with relevant news reports. Our findings show that although large state-owned and several joint-stock commercial banks began implementing digital lending platforms

in 2014 and 2015, some city and rural commercial banks followed in 2016 and afterward. Appendix Table 10 presents validation tests examining the effects of digital lending platforms on banks' digitalization performance and its constituent dimensions, using Peking University's bank digitalization index. The results show a significant increase in the overall digitalization score (*t*-value = 2.32), as well as the operational and strategic scores, from 2014, when these platforms were first introduced, to 2019, by which time all sample banks had adopted them. The operational score, which includes digital banking products like online lending, shows the strongest correlation with digital lending platforms (*t*-value = 2.23). Using the strategic digitalization score as a dependent variable, which reflects the frequency of digitalization-related terms in annual reports, the coefficient of *Digitalization* is significant at the 10% level. Using the management digitalization score as a dependent variable, which evaluates digitalization in governance and organizational processes, the coefficient of *Digitalization* is insignificant. Overall, digital lending platforms directly impact operational digitalization as operational digitalization involves integrating digital technologies into the daily operations of banking.

Main Results

In Table 2, the coefficients of *Digitalization* × *IndNPL* are negative (*t*-value = −4.46), suggesting that in industries with high NPLs, the introduction of digital lending platforms reduces loan issuance. It is important to note that since we use an *ex-ante* measure of *IndNPL*, which remains constant over the period, *IndNPL* is excluded from the regression results. Overall, the results of Table 2 support the hypothesis that compared with less digitized banks, more digitized banks provide less capital to industries with high NPLs. Economically, adopting digital lending platforms leads to a capital shift of 58% (0.029/0.05) from high-NPL industries to low-NPL industries.

Table 2 The effect of bank digitalization on capital reallocation

Dep. Var	(1) *IndProp*	(2) *IndProp*
Digitalization* × *IndNPL	**−0.029*** (−4.46)**	**−0.029*** (−4.46)**
Digitalization	0.014*** (4.49)	0.014*** (4.34)
Size		−0.002 (−1.46)
ROA		0.277 (0.81)
Lev		0.057 (1.39)
Growth		−0.007 (−1.38)
NPL		0.102 (0.73)
LLP		0.056 (0.74)
Tier1		−0.024 (−1.45)
SLR		−0.003 (−0.71)
Bank FE	YES	YES
Industry × *Year FE*	YES	YES
N	10,900	10,900
Adj R^2	0.595	0.595

This table presents the effect of bank digitalization on capital allocation. $IndProp_{i,j,t+1}$ is the proportion of loans issued to industry *j* by a specific bank *i* in year *t* + *1*. $Digitalization_{i,t}$ equals one for a specific bank *i* in a given year t if that bank has adopted the digital lending platform by that year. We use an *ex-ante* measure of industry NPLs to divide the sample into two groups. $IndNPL_{j,t}$ is assigned a value of one if the NPL ratio for industry *j* exceeds the NPL ratio in 2010, and zero otherwise. We control for bank size (*Size*), return on net assets (*ROA*), leverage ratio (*Lev*), loan growth rate (*Growth*), the proportion of non-performing loans (*NPL*), the proportion of loan loss provisions (*LLP*), capital adequacy ratio (*Tier1*), and loan-to-deposit ratio (*SLR*). Appendix Table 8 presents detailed variable definitions

The t-values are reported in parentheses near coefficients and based on standard errors clustered at the bank level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively

Heterogeneous Effects

Government Influence

In countries like China, local governments are driven by multiple objectives, including economic growth, social stability, job creation, and support for underdeveloped areas. Among these goals, achieving economic targets is of great importance. Local governments control the allocation of regional economic and administrative resources. In regions with high levels of government intervention, to bolster local economies and meet economic targets, local governments often pressure banks to allocate credit to politically favored or high-risk sectors, prioritizing political considerations over credit risk (Becker & Ivashina, 2018; Dinç, 2005; Huang & Thakor, 2024; Shang & Niu, 2023). To measure local government influence on banking, we focus on the economic goals of local officials and use the difference between actual GDP growth and target GDP growth at the provincial level as a proxy for political pressures (Cai et al., 2017). Local officials are more likely to intervene in lending decisions when they fail to achieve their economic goals. *Intervention* is defined as one for banks in provinces where the *ex-ante* gap between target and actual GDP growth rates is above the median, and zero otherwise. As shown in Panel A of Table 3, the coefficient of *Digitalization* × *IndNPL* × *Intervention* is negative at the 10% level, suggesting that the impact of bank digitalization on capital reallocation is more pronounced in regions with high levels of government influence.

Given that the sample in this study includes large state-owned commercial banks, joint-stock banks, and city and rural commercial banks, we divide the sample into two categories: national banks (comprising state-owned and joint-stock banks) and regional banks (including city and rural

Table 3 The heterogeneous impact of government influence

Panel A: Government intervention		Panel B: Bank type	
Dep. Var	*IndProp*	Dep. Var	*IndProp*
Digitalization × IndNPL × Intervention	**−0.010* (−1.73)**	***Digitalization × IndNPL × RegionalBank***	**−0.012** (−2.15)**
Digitalization	0.009* (2.00)	*Digitalization*	0.005 (1.58)
Digitalization × IndNPL	−0.018** (−2.07)	*Digitalization × IndNPL*	−0.010 (−1.66)
IndNPL × Intervention	0.023*** (2.80)	*IndNPL × RegionalBank*	0.027*** (3.73)
Digitalization × Intervention	0.004 (1.60)	*Digitalization × RegionalBank*	0.006* (2.01)
Size	−0.002 (−1.47)	*Size*	−0.002 (−1.46)
ROA	0.271 (0.79)	*ROA*	0.275 (0.80)
Lev	0.057(1.40)	*Lev*	0.057 (1.39)
Growth	−0.007 (−1.39)	*Growth*	−0.007 (−1.33)
NPL	0.102 (0.73)	*NPL*	0.101 (0.72)
LLP	0.055 (0.73)	*LLP*	0.058 (0.75)
Tier1	−0.024 (−1.53)	*Tier1*	−0.024 (−1.56)
SLR	−0.003 (−0.71)	*SLR*	−0.003 (−0.76)
Bank FE	YES	*Bank FE*	YES
Industry × Year FE	YES	*Industry × Year FE*	YES
N	10,900	*N*	10,900
Adj R^2	0.598	*Adj* R^2	0.599

This table presents the heterogeneous impact of government influence. We use the *ex-ante* difference between the target GDP growth rate and the actual GDP growth rate at the province level to measure government intervention. *Intervention* equals one for banks with an above-median government intervention and zero otherwise. *RegionalBank* is a dummy variable that equals one for rural and city commercial banks, and zero otherwise. Appendix Table 8 presents detailed variable definitions

The t-values are reported in parentheses near coefficients and based on standard errors clustered at the bank level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively

commercial banks). This allows us to examine the varying effects of digital transformation on capital reallocation across these bank types. City and rural commercial banks are often closely associated with local governments, with local government or their affiliated state-owned enterprises acting as large shareholders. This ownership structure grants local governments substantial influence over their decision-making processes. *RegionalBank* equals one for city and rural commercial banks, and zero otherwise. We incorporate the interaction between *RegionalBank* and the key explanatory variables into the regression analysis. The results, presented in Panel B of Table 3, show that the coefficient of *Digitalization × IndNPL × RegionalBank* is significantly negative at the 5% level, indicating that the impact of digital transformation on credit reallocation is more pronounced in regional banks.

Bank Transparency

Opacity in banking creates opportunities for capital misallocation by limiting the ability of regulators, investors, and auditors to monitor and evaluate lending practices effectively. Digitalization addresses this issue by enhancing transparency and increasing loan loss recognition timeliness, making banking processes more visible (Li et al., 2024; Sun et al., 2024). Consequently, stakeholders like regulators and investors have a clearer view of the bank's risk exposure. Following Akin et al. (2017), we use *ex-ante* loan loss recognition timeliness to measure banks' financial reporting quality. LLRT equals one for banks with an above-median loan loss recognition timeliness (the proportion of loan loss provisions to non-performing loans) and zero otherwise. As shown in Panel A of Table 4, the coefficient of *Digitalization × IndNPL × LLRT* is positive at the 10% level. In addition, we use *ex-ante* analyst coverage as another proxy for banks' information environments. Analyst equals one for banks with an above-median analyst coverage (the natural logarithm of one plus the number of analysts tracking the bank) and zero otherwise. The results in Table 4, Panel B, show that the coefficient of *Digitalization × IndNPL × Analyst* is 0.011, with statistical significance at the 5% level. Therefore, the impact of bank digitalization on capital reallocation is more pronounced in banks with more opaque information environments.

Internal Control and Credit Quality

Banks with weak internal control often face challenges in managing risks and are likely to engage in high-risk capital

Table 4 The heterogeneous impact of bank transparency

Panel A: Loan loss recognition timeliness		Panel B: Analyst attention	
Dep. Var	*IndProp*	Dep. Var	*IndProp*
Digitalization × IndNPL × LLRT	**0.011* (1.74)**	***Digitalization × IndNPL × Analyst***	**0.011** (2.11)**
Digitalization	0.016*** (4.97)	*Digitalization*	0.011*** (4.00)
Digitalization × IndNPL	−0.032*** (−5.02)	*Digitalization × IndNPL*	−0.023*** (−3.92)
IndNPL × LLRT	−0.005 (−0.66)	*IndNPL × Analyst*	−0.026*** (−3.79)
Digitalization × LLRT	−0.005* (−1.89)	*Digitalization × Analyst*	−0.005** (−2.13)
Size	−0.002 (−1.45)	*Size*	−0.002 (−1.47)
ROA	0.277 (0.81)	*ROA*	0.276 (0.81)
Lev	0.057 (1.37)	*Lev*	0.057 (1.40)
Growth	−0.007 (−1.37)	*Growth*	−0.007 (−1.31)
NPL	0.102 (0.75)	*NPL*	0.102 (0.73)
LLP	0.056 (0.72)	*LLP*	0.058 (0.75)
Tier1	−0.024 (−1.41)	*Tier1*	−0.024 (−1.54)
SLR	−0.003 (−0.67)	*SLR*	−0.003 (−0.74)
Bank FE	YES	*Bank FE*	YES
Industry × Year FE	YES	*Industry × Year FE*	YES
N	10,900	*N*	10,900
Adj R^2	0.595	*Adj* R^2	0.599

This table presents the heterogeneous impact of bank transparency. We use *ex-ante* loan loss recognition timeliness and analyst coverage to measure bank financial reporting quality and information environment, respectively. LLRT equals one for banks with an above-median loan loss recognition timeliness (the proportion of loan loss provisions to non-performing loans) and zero otherwise. Analyst equals one for banks with an above-median analyst coverage (the natural logarithm of one plus the number of analysts tracking the bank) and zero otherwise. Appendix Table 8 presents detailed variable definitions

The t-values are reported in parentheses near coefficients and based on standard errors clustered at the bank level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively

allocation (Jin et al., 2013). Digital lending platforms automate loan approval processes and reduce unnecessary human intervention, improving banks' internal control. These platforms also enhance real-time risk monitoring and provide ongoing oversight throughout the loan lifecycle (Doumpos et al., 2023). Thus, banks with weaker internal control benefit significantly from adopting these platforms. We use the *ex-ante* DIBO internal control index. IC equals one for banks with an above-median internal control index and zero otherwise. The results presented in Panel A of Table 5 show that bank digital transformation has a more significant effect on credit reallocation in banks with weaker internal control.

In addition, we use the *ex-ante* proportion of non-performing loans and loan concentration ratio to investigate the moderating role of credit quality. *NPL* equals one for banks with an above-median non-performing loans and zero otherwise. *Top1* equals one for banks with an above-median proportion of loan amounts concentrated in the largest clients and zero otherwise. Panels B and C of Table 5 show that the coefficients of *Digitalization × IndNPL × NPL* and *Digitalization × IndNPL × Top1* are both significantly negative at the 5% level, suggesting that the effect of digital transformation on credit reallocation is more pronounced in banks with higher credit risk. These findings reinforce the hypothesis that digital transformation is an essential tool for credit reallocation from higher-risk to lower-risk sectors, particularly in banks with low credit quality.

Robustness Tests

We conduct a series of robustness checks, as presented in Table 6. First, we test the assumption of parallel trends in Panel A. Before adopting the digital lending platform, there were no significant differences in credit reallocation trends between the treatment and control groups. However, the coefficients of *After1 × IndNPL*, *After2 × IndNPL*, *After3 × IndNPL*, and *After4 × IndNPL* are significantly negative (*t*-values = −2.39, −3.09, −3.61, and −3.06 respectively), suggesting that the proportion of loans allocated to high-NPL sectors indeed decreases after the adoption of these platforms.

Then, following Guo and Zhang (2023), this study employs the product of provincial-level internet user rates and 4G mobile phone user rates to create the instrumental variable (*Network*). In terms of relevance, this product reflects the penetration of digital infrastructure, which directly influences a bank's adoption of the digital lending

Table 5 The heterogeneous impact of internal control and credit quality

Panel A: Internal control		Panel B: Credit risk		Panel C: Loan concentration	
Dep. Var	*IndProp*	Dep. Var	*IndProp*	Dep. Var	*IndProp*
Digitalization × Ind-NPL × IC	**0.011* (2.01)**	***Digitalization × Ind-NPL × NPL***	**−0.013** (−2.45)**	***Digitalization × Ind-NPL × Top1***	**−0.011** (−2.04)**
Digitalization	0.011*** (3.60)	*Digitalization*	0.011*** (3.32)	*Digitalization*	0.008** (2.55)
Digitalization × IndNPL	−0.023*** (−3.76)	*Digitalization × IndNPL*	−0.023*** (−3.31)	*Digitalization × IndNPL*	−0.018*** (−2.75)
IndNPL × IC	−0.026*** (−4.02)	*IndNPL × NPL*	0.007 (0.86)	*IndNPL × Top1*	0.020** (2.50)
Digitalization × IC	−0.005* (−1.81)	*Digitalization × NPL*	0.006** (2.12)	*Digitalization × Top1*	0.006** (2.42)
Size	−0.002 (−1.43)	*Size*	−0.002 (−1.48)	*Size*	−0.002 (−1.45)
ROA	0.269 (0.78)	*ROA*	0.273 (0.80)	*ROA*	0.291 (0.84)
Lev	0.056 (1.37)	*Lev*	0.059 (1.40)	*Lev*	0.055 (1.37)
Growth	−0.007 (−1.27)	*Growth*	−0.007 (−1.41)	*Growth*	−0.007 (−1.39)
NPL	0.102 (0.73)	*NPL*	0.088 (0.63)	*NPL*	0.101 (0.72)
LLP	0.058 (0.76)	*LLP*	0.046 (0.62)	*LLP*	0.059 (0.78)
Tier1	−0.025 (−1.59)	*Tier1*	−0.022 (−1.38)	*Tier1*	−0.022 (−1.45)
SLR	−0.003 (−0.79)	*SLR*	−0.003 (−0.76)	*SLR*	−0.003 (−0.70)
Bank FE	YES	*Bank FE*	YES	*Bank FE*	YES
Industry × Year FE	YES	*Industry × Year FE*	YES	*Industry × Year FE*	YES
N	10,900	*N*	10,900	*N*	10,900
Adj R^2	0.598	*Adj* R^2	0.594	*Adj* R^2	0.597

This table presents the heterogeneous impact of internal control and credit quality. We use the *ex-ante* DIBO internal control index, the proportion of non-performing loans, and the loan concentration ratio. IC equals one for banks with an above-median internal control index and zero otherwise. *NPL* equals one for banks with an above-median non-performing loans and zero otherwise. *Top1* equals one for banks with an above-median proportion of loan amounts concentrated in the largest clients and zero otherwise. Appendix Table 8 presents detailed variable definitions

The t-values are reported in parentheses near coefficients and based on standard errors clustered at the bank level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively

platform. Regarding the exogeneity, mobile internet and 4G user numbers are primarily driven by technological and market conditions, with no direct relationship to a bank's loan allocation across industries, ensuring they are not correlated with the error term of the dependent variable. Thus, the *Network* is an appropriate instrumental variable. Panel B of Table 6 presents the results of the instrumental variable. Column (1) reports the first-stage regression results, where the coefficient of the *Network* is significantly positive at the 1% level, indicating the validity of this instrumental variable. The statistics at the bottom of Panel B also indicate that the *Network* is not a weak instrumental variable. Column (2) presents the second-stage regression results, the interaction term *Digitalization* × *IndNPL* remains negative at the 1% level in relation to *IndProp*, further confirming the robustness of the baseline regression results.

To address potential biases in the staggered DID analysis, we follow the approach of Barrios (2021) and Goodman-Bacon (2021) by breaking down the overall DID estimate into different group comparisons. The weight of the 2 × 2 DID estimate for the comparison between later-treated and early-treated provinces is under 10%, indicating it is not significant enough to undermine the overall DID estimate. Furthermore, we re-conduct the baseline regressions using a stacked DID design. Specifically, we calculate event-specific estimates using separate regressions for each adoption event (Cengiz et al., 2019; Gormley & Matsa, 2011). The sample period for stacked DID is from 2010 to 2017. For each adoption event before 2018, we use banks that adopted digital lending platforms in 2019 as never-treated banks and thus identify a cohort of both treated and never-treated banks. Subsequently, we pool the data across all cohorts to estimate the average treatment effect. Our model includes bank-cohort and industry × year-cohort fixed effects (Gormley & Matsa, 2011). The significant negative coefficient of *Stacked Digitalization* × *IndNPL* (Coefficient = −0.048, $t = -4.31$) indicates that bank digitalization facilitates the capital reallocation from sectors with a higher proportion of NPLs to those with a lower proportion of NPLs.

In Panel D, we employ an alternative measure of bank digitalization. Specifically, we utilize Peking University's bank digital transformation index (*Digi*) and its three sub-components—strategic digitalization (*Strategic*), operational digitalization (*Operational*), and management digitalization (*Management*)—as independent variables. The results in Panel D indicate that the coefficients of all dimensions of

Table 6 Robustness tests

Panel A: Parallel trend assumption

Dep. Var	*IndProp*
Before2 × IndNPL	0.002 (1.24)
Before1 × IndNPL	−0.000 (−0.15)
After1 × IndNPL	**−0.004** (−2.39)**
After2 × IndNPL	**−0.006*** (−3.09)**
After3 × IndNPL	**−0.008*** (−3.61)**
After4 × IndNPL	**−0.013*** (−3.06)**
Controls	YES
Bank FE	YES
Industry × Year FE	YES
N	10,900
Adj R^2	0.593

Panel B: Instrumental variable

Dep. Var	(1)	(2)
	Digitalization	*IndProp*
Network	**0.078*** (3.01)**	
Digitalization × IndNPL		**−0.085*** (2.63)**
Controls	YES	YES
Bank FE	YES	YES
Industry × Year FE	YES	YES
Kleibergen–Paap rk LM test	9.91 (0.007)	
Cragg–Donald Wald F test	367.36 (19.93)	
Anderson–Rubin Wald test	15.24 (0.001)	
N	10,900	10,900
Adj R^2	0.591	0.591

Panel C: Stacked DID

Dep. Var	*IndProp*
Stacked Digitalization × IndNPL	**−0.048*** (−4.31)**
Stacked Digitalization	0.023*** (3.88)
Controls	YES
Bank-Cohort FE	YES
Industry × Year-Cohort FE	YES
N	8820
Adj R^2	0.647

Panel D: Alternative measure of bank digitalization

Dep. Var	(1)	(2)	(3)	(4)
	IndProp			
Digi × IndNPL	**−0.001*** (−4.55)**			
Strategic × IndNPL		**−0.001*** (−3.91)**		
Operational × IndNPL			**−0.001*** (−4.49)**	
Management × IndNPL				**−0.001** (−2.78)**
Controls	YES	YES	YES	YES
Bank FE	YES	YES	YES	YES
Industry × Year FE	YES	YES	YES	YES
N	9560	9560	9560	9560

Table 6 (continued)

Panel D: Alternative measure of bank digitalization				
Dep. Var	(1)	(2)	(3)	(4)
	IndProp			
Adj R^2	0.617	0.616	0.617	0.613

Panel E: Alternative sample		
	(1)	(2)
Dep. Var	Deletion of developed cities	Deletion of the post-2019 period
	IndProp	
Digitalization × IndNPL	**−0.026*** (−3.12)**	**−0.029*** (−4.46)**
Digitalization	0.012*** (2.86)	0.014*** (4.22)
Controls	YES	YES
Bank FE	YES	YES
Industry × Year FE	YES	YES
N	7260	8380
Adj R^2	0.598	0.592

This table presents the results of the robustness tests. In Panel A, we use the digital lending platform launch year as the benchmark. Panels B and C apply instrumental variable and stacked DID analyses, respectively. In Panel B, the weak IV test employs the Cragg–Donald Wald F statistic, with the critical value at the 10% level provided in parentheses, as per Stock-Yogo. The under-identification and exogeneity tests are conducted using the Kleibergen–Paap rk LM statistic and the Anderson–Rubin Wald statistic, respectively, with the corresponding *p* values reported in parentheses. In Panel D, we use the bank digitalization index developed by Peking University, which includes three dimensions—strategy, operations, and management—as alternative measures of bank digitalization. In Panel E, we exclude samples from developed cities (Beijing, Shanghai, Guangzhou, and Shenzhen) and omit data from the post-2019 period to mitigate the potential effects of the COVID-19 pandemic on capital allocation. Appendix Table 8 presents detailed variable definitions

The t-values are reported in parentheses near coefficients and based on standard errors clustered at the bank level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively

digitalization are statistically significant, consistent with our primary findings.

Lastly, to mitigate concerns that highly digitized banks are concentrated in economically developed cities, we exclude four megacities (Beijing, Shanghai, Guangzhou, and Shenzhen) in column (1) of Panel E, the coefficient of *Digitalization × IndNPL* remains significant (*t*-value = −3.12). Additionally, to account for the potential impact of the COVID-19 pandemic on bank credit reallocation, we exclude the post-2019 sample and re-run the regressions. As reported in column (2) of Panel E, the results are still robust (*t*-value = −4.46).

Further Analyses

Loan Loss Recognition Timeliness

Before the adoption of digital lending platforms, the recognition of loan loss provisions often depends on the subjective judgment of credit officers and risk managers, leading to potential delays, data inconsistencies, and slow decision-making. In contrast, digital lending platforms improve this process by using real-time monitoring and risk assessment, which reduces delays and errors associated with manual methods (Sun et al., 2024). Digital lending platforms utilize data analytics to integrate diverse data sources and perform advanced risk evaluations. By analyzing borrower behavior, market conditions, and macroeconomic factors, digital lending platforms identify existing defaults more quickly and predict expected credit risks. This predictive capability enhances banks' ability to manage emerging risks proactively and improves the timeliness of loan loss provisions.

To evaluate the impact of digitalization on bank transparency, we regress loan loss recognition timeliness in the next period on banks' adoption of digital lending platforms. As shown in column (1) of Table 7, *Digitalization* is positive at the 10% level, suggesting that as banks advance in their digital transformation, the timeliness of loan loss provisions improves. Utilizing data analytics and comprehensive risk management systems, banks are likely to detect loan losses more efficiently and thus increase loan loss recognition timeliness, providing regulators and investors with a transparent view of banks' risk exposure.

Table 7 Further analyses

Dep. Var	(1) *LLRT*	(2) *Z-Score*	(3) *NetProfit*
Digitalization	**0.014* (1.97)**	**0.241* (1.89)**	**0.016*** (2.91)**
Size	−0.045*** (−5.46)	0.017 (0.47)	−0.001 (−0.10)
ROA	1.530 (0.58)	−5.525 (−0.93)	11.197*** (5.48)
Lev	1.259*** (5.23)	−0.336 (−0.35)	−0.098 (−0.26)
Growth	0.113* (1.85)	0.310 (1.19)	0.046 (1.06)
NPL	−14.685*** (−8.78)	4.677 (0.82)	−5.018*** (−4.87)
LLP	7.931*** (9.52)	−4.637 (−1.23)	1.194** (2.30)
Tier1	0.592** (2.39)	−2.608 (−1.05)	0.058 (0.56)
SLR	−0.062 (−1.28)	0.269 (1.58)	−0.022 (−0.51)
Bank FE	YES	YES	YES
Year FE	YES	YES	YES
N	9920	8880	9800
Adj R^2	0.795	0.154	0.759

This table delves deeper into the welfare implications of bank digitization. *LLRT* is the proportion of the loan loss provisions to non-performing loans. *Z-Score* assesses banks' risk, which is defined as the sum of ROA and the capital adequacy ratio, divided by the standard deviation of ROA. *NetProfit* is the net profit divided by revenue. Appendix Table 8 presents detailed variable definitions

The t-values are reported in parentheses near coefficients and based on standard errors clustered at the bank level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively

Bank Risk

From the perspective of bank risk, we examine the welfare implications of reallocating capital from higher-risk to lower-risk sectors and how this shift impacts banks' idiosyncratic risk. Digital transformation results in data-driven credit decision-making (Wang et al., 2024) and enhances banks' ability to respond to evolving risks by allowing for real-time risk assessments (Butler & Brien, 2019). Digital lending platforms incorporate post-lending monitoring systems that continuously track risk developments and reassess credit limits. When anomalies are detected, automated alerts notify the bank's risk management team and clients, prompting necessary actions such as adjusting loan limits, increasing collateral requirements, requesting additional margin deposits, or implementing risk mitigation measures. Utilizing the next-period *Z-Score* as a proxy for bank risk exposure, our regression analysis, as presented in column (2) of Table 7, indicates that *Digitalization* is significant at the 10% level (t-value = 1.89). This further substantiates the positive role of digital transformation in enhancing bank risk management.

Operating Performance

On the one hand, capital reallocation from high-risk to low-risk sectors brought by digitalization may reduce banks' profitability. On the other hand, bank digital transformation helps lower operational costs. Digital lending platforms streamline traditional credit approval procedures by minimizing manual interventions. This leads to a substantial reduction in labor, time, and credit processing costs, allowing banks to extend credit at lower costs. The real-time data analytics and automated approval processes accelerate loan disbursement and enhance operational efficiency (Wang et al., 2021). Fuster et al. (2019) provide evidence that fintech lenders respond to mortgage demand shocks more efficiently than traditional lenders, processing applications more quickly and adjusting supply with greater flexibility, indicating that technological innovation can enhance the efficiency of financial intermediation in the mortgage market. Buchak et al. (2018) highlight that Quicken Loans' Rocket Mortgage product, which operates almost entirely online without the involvement of human loan officers, reduces both labor and office costs, while significantly increasing customer convenience. We use the proportion of net profit to total revenue (*NetProfit*) as a proxy for bank profitability. The results, displayed in column (3) of Table 7, show that adopting digital lending platforms has enhanced banks' profitability (t-value = 2.91).

Conclusions

Extant studies have explored the impact of digitalization on the operation, productivity, equity values, and overall performance of non-financial firms (Chen & Srinivasan, 2023; Correani et al., 2020). Our paper, however, switches the attention to bank digitalization. While a small number of studies have examined the effects of digital transformation

on systemic risk, credit provision to SMEs, and overall bank performance (Chen & Shen, 2024; Cheng & Qu, 2020; Sheng, 2021; Zhao et al., 2022), there is still limited understanding of how bank digitalization influences capital allocation across different sectors. This study highlights the disciplinary role of digitalization in facilitating capital reallocation from higher-risk to lower-risk sectors.

Our findings indicate that bank digitalization can effectively redirect capital from sectors with higher NPLs to those with lower NPLs by reducing political pressures and improving transparency and risk management. Banks that adopt digital lending platforms are better equipped to resist external interference, as the data-driven, automated nature of decision-making makes it harder to justify risky, politically motivated loans. This shift concentrates in regions with high levels of government interference, as well as in city and rural commercial banks, which are more susceptible to government intervention. Given that banks with opaque information environments and weak internal control face significant challenges in managing risk and ensuring the accuracy of their financial reporting, we find that the impact of bank digitalization is more pronounced in these banks with opaque information environments and weak internal control. Further analyses reveal that bank digitalization increases loan loss recognition timeliness, reduces bank risk, and increases operating performance. While our study focuses on China's banking system, our findings are relevant to other countries as well. Many emerging economies face similar issues like government intervention in banking, inefficient capital allocation, and risky lending practices. In places where political influence affects credit decisions, digital lending platforms can help increase transparency and reduce unnecessary interference. Because these platforms rely on data-driven, automated decision-making, it is harder to justify loans based on non-economic reasons. This promotes more efficient and ethical capital allocation across different institutional settings.

However, our study has certain limitations. Our study focuses on the impact of digital lending platforms on bank capital allocation. This focus may not cover all aspects of digitalization within banks. Other areas of digital transformation, such as mobile banking and digital customer service, are not included in our analysis. Future research could investigate these additional dimensions to better understand how digitalization affects banking operations and risk management. Additionally, our study is based on listed banks due to limited data on non-listed banks' adoption of digital lending platforms. Future research could address this limitation by including a broader range of banks to enhance the generalizability of the findings.

Overall, this paper underscores the critical impact of digital transformation as a powerful tool in fostering a more ethical and efficient banking system. It has important consequences for the stability of China's financial system, particularly in regions with high levels of government intervention and banks with weaker internal control and less transparency. Ethically, the movement toward digital lending platforms supports a more transparent financial environment, where transactions and credit decisions are open and accountable. This promotes a banking culture that values fairness and transparency, crucial for ethical business practices. Moreover, by enhancing transparency and accountability, digital platforms offer regulators a powerful tool to improve governance and address the bank risks posed by political intervention and high-risk lending.

Banks should continue to advance their digital transformation efforts, leveraging advanced data analytics and risk management tools to optimize credit allocation and reduce credit risks. Regulators should support these initiatives by creating a regulatory environment that encourages digital innovation, while ensuring robust oversight. This support includes updating regulatory frameworks to address new risks introduced by digital technologies and promoting standards that ensure data privacy and security. Policymakers should prioritize digitalization to enhance individual bank performance and support a more stable and resilient financial system.

Appendix

See Tables 8, 9, and 10.

Table 8 Variable definitions

Variable	Definition	Data Source
Dependent variable		
IndProp	The proportion of loan issued to industry *j* by a specific bank *i* in year *t*	The annual report of the Bank Regulatory Commission
Independent variables		
IndNPL	An indicator variable that equals one if the proportion of NPL for industry *j* exceeds the median NPL ratio in 2010, and zero otherwise	The annual report of the Bank Regulatory Commission
Digitalization	An indicator variable that equals one for a specific bank *i* in a given year t if that bank has adopted the digital lending platform by that year	Company annual report
Digi	Bank digitalization level evaluated from three dimensions including strategic, operation, and management	Commercial Bank Digital Transformation Index in China by Peking University
Strategic	Number of mentions of six types of digital technology keywords in the annual report: AI-related, blockchain-related, cloud computing-related, big data-related, online-related, and mobile-related	Commercial Bank Digital Transformation Index in China by Peking University
Operational	The three dimensions of digital channels, digital products, and digital R&D	Commercial Bank Digital Transformation Index in China by Peking University
Management	The three dimensions of digital architecture, digital talent, and digital partnerships	Commercial Bank Digital Transformation Index in China by Peking University
Control variables		
Size	The logarithm of total assets	Wind
Lev	Total liabilities/Total assets	Wind
NPL	Non-performing loans/Gross loans	Wind
LLP	Loan loss provisions/Total loans	Wind
SLR	Total loans/Total deposits	Wind
Tier1	Tier 1 capital/Risk-weighted assets	Wind
ROA	Net profit/Total assets	Wind
Growth	(End-of-period total loans − Beginning-of-period total loans)/Beginning-of-period total loans	Wind
Additional variables		
Intervention	The difference between the target GDP growth rate and actual GDP growth rate at the province level	Statistical yearbook
LLRT	The proportion of the loan loss provisions to non-performing loans	CSMAR
Top1	The proportion of loan amounts from the top client relative to the total loans	CSMAR
Z-Score	(ROA + Equity/Assets)/Standard deviation of ROA	CSMAR
RegionalBank	An indicator variable that equals one for rural commercial bank and city commercial bank, and zero otherwise	Wind
Analyst	The natural logarithm of one plus the number of analysts tracking the company	CSMAR
IC	The natural logarithm of internal control index	DIB database
NetProfit	Net profit/revenue	CSMAR

Table 9 The adoption time of digital lending platforms

Bank name	Bank type	Year
China Construction Bank	Large State-Owned Commercial Bank	2014
China CITIC Bank	Joint-Stock Commercial Bank	2015
China Everbright Bank	Joint-Stock Commercial Bank	2015
Industrial and Commercial Bank of China	Large State-Owned Commercial Bank	2015
Bank of China	Large State-Owned Commercial Bank	2015
Shanghai Pudong Development Bank	Joint-Stock Commercial Bank	2015
Bank of Communications	Large State-Owned Commercial Bank	2015
China Minsheng Bank	Joint-Stock Commercial Bank	2016
Industrial Bank	Joint-Stock Commercial Bank	2016
Bank of Hangzhou	City Commercial Bank	2016
Bank of Jiangsu	City Commercial Bank	2016
Bank of Guiyang	City Commercial Bank	2016
Agricultural Bank of China	Large State-Owned Commercial Bank	2016
Bank of Nanjing	City Commercial Bank	2016
Bank of Ningbo	City Commercial Bank	2016
Bank of Chongqing	City Commercial Bank	2016
Zheshang Bank	Joint-Stock Commercial Bank	2016
Postal Savings Bank of China	Large State-Owned Commercial Bank	2017
China Merchants Bank	Joint-Stock Commercial Bank	2017
Ping An Bank	Joint-Stock Commercial Bank	2017
Bank of Xi'an	City Commercial Bank	2017
Bank of Lanzhou	City Commercial Bank	2017
Bank of Zhengzhou	City Commercial Bank	2017
Bank of Suzhou	City Commercial Bank	2017
Bank of Chengdu	City Commercial Bank	2018
Xiamen Bank	City Commercial Bank	2017
Bank of Beijing	City Commercial Bank	2018
Jiangsu Jiangyin Rural Commercial Bank	Rural Commercial Bank	2018
Jiangsu Zijin Rural Commercial Bank	Rural Commercial Bank	2018
Chongqing Rural Commercial Bank	Rural Commercial Bank	2018
Bank of Shanghai	City Commercial Bank	2018
Bank of Changsha	City Commercial Bank	2018
Huaxia Bank	Joint-Stock Commercial Bank	2018
Qilu Bank	City Commercial Bank	2018
Qingdao Rural Commercial Bank	Rural Commercial Bank	2018
Jiangsu Suzhou Rural Commercial Bank	Rural Commercial Bank	2019
Bank of Qingdao	City Commercial Bank	2019
Jiangsu Changshu Rural Commercial Bank	Rural Commercial Bank	2019
Jiangsu Wuxi Rural Commercial Bank	Rural CommercialBank	2019
Jiangsu Zhangjiagang Rural Commercial Bank	Rural Commercial Bank	2019
Shanghai Rural Commercial Bank	Rural Commercial Bank	2019
Shaoxing Ruifeng Rural Commercial Bank	Rural Commercial Bank	2019

Table 10 Validation tests: the adoption of digital lending platforms and digitalization performance

Dep. Var	(1) *Digi*	(2) *Operational*	(3) *Strategic*	(4) *Management*
Digitalization	**0.086** (2.32)**	**0.095** (2.23)**	**0.247* (1.90)**	**0.031 (0.79)**
Lev	−0.091 (−0.05)	−8.084*** (−3.37)	6.141 (0.85)	3.067 (1.56)
Size	0.001 (0.02)	0.262*** (3.14)	−0.210 (−0.82)	−0.101 (−1.42)
Growth	−0.268 (−1.16)	0.190 (0.59)	−0.683 (−0.77)	−0.416 (−1.48)
ROA	12.474 (1.20)	39.388* (1.96)	25.589 (0.60)	−7.990 (−0.54)
Bank FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
N	4100	4100	4100	4100
Adj R^2	0.867	0.771	0.808	0.758

The t-values are reported in parentheses near coefficients and based on standard errors clustered at the bank level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively

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Data availability Data is available upon request.

Declarations

Conflicts of interest The authors declare that they have no financial or non-financial interests that are directly or indirectly related to the research described in this paper.

Research Involving Human Participants and/or Animals N/A.

Informed Consent N/A.

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